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Nirma University

Institute of Management

Trimester End Examination (IR/RPR), October-2022

Master of Business Administration, Trimester - IV (makeup)

MFT5SEQ06 Lean Six Sigma

Roll / Exam No.

Supervisor's Initial
with Date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Attempt questions in the same sequence
5. Attach the question paper with the answer sheet

Q1. A government agency contracted with a company to design and build a satellite system. Months after the contract was signed, the company discovered that the design would not be immune to certain types of radar interference. The agency claimed that it had described the performance desired for the satellites. The company disagreed (with respect to the radar interference). If this need had been realized at the start of the project, creating an appropriate design would have been relatively simple. The satellites are in an advanced stage of design and construction, and the necessary changes would cost \$100 million. There was further confusion. The company had chosen a supplier to manufacture the satellites. This supplier had previous experience with such products, and some people claimed that the supplier should have been aware of the radar interference problem. Comment on the actions that should be taken by three organizations to prevent such a situation on a future project.

(10 Marks)

Q2. How the Achieving Competitive Excellence (ACE) as implemented in the Eagle Services Asia differs from the ACE (Apollo Clinical Excellence)?

(10 Marks)

Q3. What is Muda, Mura and Muri concept? Why it is so hard to implement the same?

(10 Marks)

Q4. Consider a cell phone manufacturer who wants to calculate the six-sigma level of its manufacturing process. For a given period, the manufacturer makes 83,934 cell phones. The

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manufacturer is required to perform eight checks to test the quality of its products. During testing phase, 3,432 of the cell phones were rejected.

(10 Marks)

Q5. What is Single Minute Exchange of Dies concept? Who introduced it? Quote and explain at least three examples where this concept could be put to use.

(10 Marks)

Q6. Which objections might a marketing manager have against uniform plant loading?

(10 Marks)

Q7. Why is lean so hard to implement in practice? In what type of environment is lean production most successful?

(10 Marks)

Read the attached case "Copper Kettle..." and answer the following three questions:

Q8. Are the operations of Copper Kettle Catering conducive to the application of JIT concepts and practices? Explain.

(10 Marks)

Q9. What, if any, are the major barriers to implementing a JIT system at Copper Kettle Catering?

(10 Marks)

Q10. What would you recommend that Wayne and Janet Williams do to take advantage of JIT concepts in operating CKC?

(10 Marks)

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CASE COPPER KETTLE CATERING

Copper Kettle Catering (CKC) is a full-service catering company that provides services ranging from box lunches for picnics or luncheon meetings to large wedding, dinner, or office parties. Established as a lunch delivery service for offices in 1972 by Wayne and Janet Williams, CKC has grown to be one of the largest catering businesses in Raleigh, North Carolina. The Williamses divide customer demand into two categories: *deliver only* and *deliver and serve*.

The deliver-only side of the business provides dropoff of boxed meals consisting of a sandwich, salad, dessert, and fruit. The menu for this service is limited to six sandwich selections, three salads or potato chips, and a brownie or fruit bar. Grapes and an orange slice are included with every meal, and iced tea can be ordered to accompany the meals. The overall level of demand for this service throughout the year is fairly constant, although the mix of menu items delivered varies. The planning horizon for this segment of the business is short: Customers usually call no more than a day ahead of time. CKC requires customers to call deliver-only orders in by 10:00 A.M. to guarantee delivery the same day.

The deliver-and-serve side of the business focuses on catering large parties, dinners, and weddings. The extensive range of menu items includes a full selection of hors d'oeuvres, entrées, beverages, and special-request items. The demand for these services is much more seasonal, with heavier demands occurring in the late spring-early summer for weddings and the late fall-early winter for holiday parties. However, this segment also has a longer planning horizon. Customers book dates and choose menu items weeks or months ahead of time.

Copper Kettle Company's food preparation facilities support both operations. The physical facilities layout resembles that of a job shop. There are five major work areas: a stove-oven area for hot food preparation, a cold area for salad preparation, an hors d'oeuvre preparation area, a sandwich preparation area, and an assembly area where deliver-only orders are boxed and deliver-and-serve orders are assembled and trayed. Three walk-in coolers store foods requiring refrigeration, and a large pantry houses nonperishable goods. Space limitations and the risk of spoilage limit the amount of raw materials and prepared food items that can be carried in inventory at any one time. CKC purchases desserts from outside vendors. Some deliver the desserts to CKC; others require CKC to send someone to pick up desserts at their facilities.

The scheduling of orders is a two-stage process. Each Monday, the Williamses develop the schedule of deliver-and-serve orders to be processed each day. CKC typically has multiple deliver-and-serve orders to fill each day of the week. This level of demand allows a certain efficiency in

preparation of multiple orders. The deliver-only orders are scheduled day to day owing to the short-order lead times. CKC sometimes runs out of ingredients for deliver-only menu items because of the limited inventory space.

Wayne and Janet Williams have 10 full-time employees: two cooks and eight food preparation workers, who also work as servers for the deliver-and-serve orders. In periods of high demand, the Williamses hire additional part-time servers. The position of cook is specialized and requires a high degree of training and skill. The rest of the employees are flexible and move between tasks as needed.

The business environment for catering is competitive. The competitive priorities are high-quality food, delivery reliability, flexibility, and cost—in that order. "The quality of the food and its preparation is paramount," states Wayne Williams. "Caterers with poor-quality food will not stay in business long." Quality is measured by both freshness and taste. Delivery reliability encompasses both on-time delivery and the time required to respond to customer orders (in effect, the order lead time). Flexibility focuses on both the range of catering requests that a company can satisfy and menu variety.

Recently, CKC has begun to feel the competitive pressures of increasingly demanding customers and several new specialty caterers. Customers are demanding more menu flexibility and faster response times. Small specialty caterers have entered the market and have targeted specific well-defined market segments. One example is a small caterer called Lunches-R-US, which located a facility in the middle of a large office complex to serve the lunch trade and competes with CKC on cost.

Wayne and Janet Williams have been impressed by the concepts of just-in-time operating systems, especially the ideas of increasing flexibility, reducing lead times, and lowering costs. They sound like what CKC needs to do to remain competitive. But the Williamses wonder whether JIT concepts and practices are transferable to a service business.

Questions

1. Are the operations of Copper Kettle Catering conducive to the application of JIT concepts and practices? Explain.
2. What, if any, are the major barriers to implementing a JIT system at Copper Kettle Catering?
3. What would you recommend that Wayne and Janet Williams do to take advantage of JIT concepts in operating CKC?

Source: This case was prepared by Dr. Brooke Saladin, Wake Forest University, as a basis for classroom discussion.